

API QA Test Report

TaskFlow REST API · Postman, Newman and pytest verification · Release candidate 1.0

Portfolio disclosure
TaskFlow is a fictional QA target built for demonstration. The historical defects below are seeded examples, not claims about a real client.

12 REQUESTS EXECUTED	41 POSTMAN ASSERTIONS	0 FAILED ASSERTIONS	11 / 11 PYTEST CHECKS PASSED	8.8 ms AVG. LOCAL RESPONSE
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Objective Verify core API behavior, error handling, response contracts and a create-to-delete workflow before release. Coverage Health, login, bearer authorization, validation, pagination, user lookup, task creation and deletion.	Release recommendation PASS — suitable for demonstration release. All 41 Newman assertions and 11 pytest regression checks passed. No open seeded defects remain in this version.
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Test Coverage Matrix

Status and expected behavior for each request in the recorded Newman run.

ID	Area	Method	Endpoint	Scenario	Expected result	Type	Status
TF-001	System	GET	/health	Service health check	200 with status and version	Smoke	Pass
TF-002	Authentication	POST	/auth/login	Valid credentials	200 with bearer token	Positive	Pass
TF-003	Authentication	POST	/auth/login	Invalid password	401 with explicit error	Negative	Pass
TF-004	Authentication	POST	/auth/login	Missing password	422 with validation detail	Negative	Pass
TF-005	Authorization	GET	/users	Missing bearer token	401 with token error	Negative	Pass
TF-006	Users	GET	/users?page=1&page_size=2	Valid pagination	200 with two users and metadata	Positive	Pass
TF-007	Users	GET	/users?page=0	Invalid lower boundary	422 with boundary error	Boundary	Pass
TF-008	Users	GET	/users/1	Existing user	200 with expected user contract	Positive	Pass
TF-009	Users	GET	/users/999	Unknown user	404 with explicit error	Negative	Pass
TF-010	Tasks	POST	/tasks	Valid task	201 with preserved values	Positive	Pass
TF-011	Tasks	POST	/tasks	Title below minimum length	422 with validation detail	Boundary	Pass
TF-012	Tasks	DELETE	/tasks/101	Delete existing task	204 with empty body	Workflow	Pass

Defect Verification

Seeded defect history shows how findings are prioritized, communicated and retested after a fix.

ID	Severity	Area	Finding	Previous behavior	Expected behavior	Resolution	Verification
QA-001	High	Authentication	Missing password caused server error	POST /auth/login returned 500	Return a client validation error	Fixed in demo v1.0	Retested: 422 Pass
QA-002	Medium	Users	Page zero was accepted	GET /users?page=0 returned 200	Reject invalid pagination boundary	Fixed in demo v1.0	Retested: 422 Pass
QA-003	Medium	Users	Unknown user returned null	GET /users/999 returned 200 with null	Return explicit not-found response	Fixed in demo v1.0	Retested: 404 Pass
QA-004	Low	Tasks	Created task used wrong status	POST /tasks returned 200	Return 201 Created	Fixed in demo v1.0	Retested: 201 Pass

High Core workflow or reliability risk	Medium Incorrect behavior with a viable workaround	Low Contract or standards inconsistency
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Method and Deliverables

1	Understand the contract	Review Swagger/OpenAPI, authentication, environments, business rules and success criteria.
2	Design coverage	Map happy paths, negative paths, missing/invalid fields, boundaries, status codes and response schemas.
3	Execute and automate	Build an organized Postman collection, capture variables, write assertions and execute with Newman.
4	Report findings	Log severity, exact reproduction steps, expected/actual behavior, evidence and business impact.
5	Verify fixes	Rerun affected cases and regression coverage, then provide a release recommendation.

Included files Postman collection and environment Test matrix Defect log Newman JSON evidence pytest suite README and CI workflow	Demonstrated skills REST API QA Authentication and authorization Negative and boundary testing JSON contract checks Postman and Newman pytest, FastAPI and CI	Scope boundary This report demonstrates functional API QA. It is not a penetration test, load test, production audit or guarantee of defect-free software.
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Evidence integrity: counts in this report are read directly from report s/newman-run.json; the source API and all automated tests are included with the portfolio.